

```
In [1]: pip install pandas
```

Requirement already satisfied: pandas in C:\Users\afgha\anaconda3\Lib\site-pa (3.0.3)
Requirement already satisfied: numpy>=2.3.3 in C:\Users\afgha\anaconda3\Lib\s packages (from pandas) (2.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\afgha\anaconda3\Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\afgha\anaconda3\Lib\site-pa (from pandas) (2026.2)
Requirement already satisfied: six>=1.5 in C:\Users\afgha\anaconda3\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

```
In [2]: import pandas as pd  
data = pd.read_csv("trends.csv")
```

```
In [3]: data.head()
```

```
Out[3]:
```

	location	year	category	rank	query
0	Global	2 0 0 1	Consumer Brands	1	Nokia
1	Global	2 0 0 1	Consumer Brands	2	Sony
2	Global	2 0 0 1	Consumer Brands	3	BMW
3	Global	2 0 0 1	Consumer Brands	4	Palm
4	Global	2 0 0 1	Consumer Brands	5	Adobe

```
In [4]: data.describe()
```

```
Out[4]:
```

	year	rank
count	2 6 9 5 5.0 0 0 0 0 0	2 6 9 5 5.0 0 0 0 0
mean	2 0 1 5.2 4 3 3 6 9	3.0 0 0 0 0 0
std	3.5 6 4 6 8 3	1.4 1 4 2 4
min	2 0 0 1.0 0 0 0 0 0	1.0 0 0 0 0 0
2 5 %	2 0 1 3.0 0 0 0 0 0	2.0 0 0 0 0 0
5 0 %	2 0 1 6.0 0 0 0 0 0	3.0 0 0 0 0 0
7 5 %	2 0 1 8.0 0 0 0 0 0	4.0 0 0 0 0 0
max	2 0 2 0.0 0 0 0 0 0	5.0 0 0 0 0 0

```
In [5]: data.shape
```

```
Out[5]: (26955, 5)
```

```
In [6]: data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 26955 entries, 0 to 26954
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   location    26955 non-null  str
1   year        26955 non-null  int64
2   category    26955 non-null  str
3   rank        26955 non-null  int64
4   query       26955 non-null  str
dtypes: int64(2), str(3)
memory usage: 2.0 MB
```

```
In [7]: data.columns
```

```
Out[7]: Index(['location', 'year', 'category', 'rank', 'query'], dtype='str')
```

```
In [8]: data.isnull().sum()
```

```
Out[8]: location    0
year              0
category          0
rank              0
query             0
dtype: int64
```

```
In [10]: Q1 = data['rank'].quantile(0.25)
Q3 = data['rank'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = data[(data['rank'] < lower_bound) | (data['rank'] > upper_bound)]
```

```
In [11]: print(outliers)
```

```
Empty DataFrame
Columns: [location, year, category, rank, query]
Index: []
```

```
In [12]: print(lower_bound)
```

```
-1.0
```

```
In [13]: print(upper_bound)
```

```
7.0
```

```
In [14]: data.nunique()
```

```
Out[14]: location      83
year                20
category           2450
rank                5
query             18431
dtype: int64
```

```
In [15]: data.duplicated().sum()
```

```
Out[15]: np.int64(10)
```

```
In [17]: data.drop_duplicates(inplace=True)
```

```
In [18]: data.info()
```

```
<class 'pandas.DataFrame'>  
Index: 26945 entries, 0 to 26954  
Data columns (total 5 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   location    26945 non-null  str  
1   year        26945 non-null  int64  
2   category    26945 non-null  str  
3   rank        26945 non-null  int64  
4   query       26945 non-null  str  
dtypes: int64(2), str(3)  
memory usage: 2.2 MB
```

```
In [ ]:
```

```
In [ ]:
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